Smartphone Recommendation System

Content-Based | Utility-Based | Hybrid

Group 6 — Recommender Systems Mini-Project

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Problem & Objectives

Help users find smartphones matching budget and feature priorities.

Input: Budget (BDT), priorities, optional reference phone.

Output: Ranked top-five eligible smartphones.

Dataset: Real Kaggle 2026

Global Mobile Phone Specifications & Prices (UTSHO DEY).
6,620 source records and 168 columns.
4,627 screened smartphone records after filtering and cleaning.
Price currency: Bangladeshi taka (BDT), not USD/SAR.

Preprocessing

Select smartphones and mobile devices only.
Parse numeric RAM, storage, battery, camera and screen features.
Exclude implausible or feature-phone-like hardware values; retain original CSV.
Remove unusable prices and duplicate models.
Rescale ratings and apply median imputations / Min–Max scaling.

Method 1: Content-Based

Cosine similarity between smartphone feature vectors and user priority target.

Uses RAM, storage, battery, camera, screen size and rating.

Method 2: Utility-Based

Q = weighted feature quality; A = max(0, 1 − price/budget).

Utility = 0.8 Q + 0.2 A.

Gaming, camera, battery and balanced priority profiles.

Method 3: Hybrid

Hybrid = α × Content score + (1 − α) × Utility score

Default α = 0.5; adjustable in the Streamlit interface.

Budget constraints apply before ranking.

Results: 30,000 BDT Gaming

1. Xiaomi Redmi Turbo 3 — BDT 28,000, score 0.689
2. Xiaomi Redmi Note 12R Pro — BDT 28,000, score 0.682
3. T-Mobile REVVL 7 Pro — BDT 15,000, score 0.669
4. Oscal Pilot 3 — BDT 30,000, score 0.669
5. Motorola Edge 60s — BDT 30,000, score 0.667

Evaluation & Comparison

Eligible smartphones: 2,802 / 4,627.
Content–Utility overlap@5: 0.00.
Hybrid top-five mean listed price: BDT 26,200.
No user relevance ground truth: these scores are NOT measured accuracy.

Demo, Conclusions & Limitations

Live demo: streamlit run app.py → budget 30000 → priority gaming → hybrid.
Three working recommender methods, now operating on a real dataset.
Limitations: historical BDT prices, parsed specs, hardware proxy scores, no user relevance labels.
Future: verified local prices and prospective user preference tests.